Transparency without pricing: a credit-level forensic account of collapse in the first tokenized carbon pool

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hen assets of different quality are pooled and sold at a single price, economic theory predicts that low-quality units drive out high-quality ones — but, classically, only when buyers cannot tell the difference. We show that this collapse also occurs when quality is fully and publicly observable, if the market's design does not let price reflect quality. We test this in the largest pooled carbon-credit market (22 million tonnes of CO_2), whose decline was widely blamed on low-quality forest credits. Analysing every deposit, withdrawal, and participant account recorded on its public ledger, we find the opposite: the pool held 69% renewable-energy credits with little additional climate benefit and only 4% forest credits. Two almost separate groups operated on each side — sellers deposited indiscriminately, while a few accounts removed most of the high-value credits (1.55 million tonnes), and 9.6 million tonnes of low-value credits are now permanently stranded. Tracking nature-based credits that entered both an unscreened and a quality-screened pool, the same credits were almost fully redeemed from the unscreened pool but largely retained in the screened one — a contrast strongly consistent with pool design, rather than credit type, shaping their fate. Transparency alone therefore appears insufficient to prevent quality collapse: pooled markets may need to price quality, not merely disclose it — a caution for carbon policy and for the $10-billion market now tokenizing other real-world assets on the same pooled design.

Introduction

Every year, companies and governments purchase carbon credits worth billions of dollars to offset their greenhouse gas emissions. A carbon credit represents one tonne of CO_2 prevented from entering the atmosphere, but the climate value of individual credits varies enormously. The central question for any carbon market is whether it can distinguish credits that represent genuine emission reductions from those that do not.

When market design fails to price quality differences, the result is adverse selection — a dynamic in which low-quality goods crowd out high-quality ones, first formalized by Akerlof in 1970 as the "lemons problem"[1,2,3]. The voluntary carbon market (VCM), valued at over $2 billion annually, faces exactly this risk. Fewer than 16% of credits from investigated projects represent genuine reductions[4]. At least 52% of Indian wind power offsets went to projects that would have been built regardless of carbon revenue[5].

Blockchain-based tokenization promised to solve this problem through radical transparency. By recording every transaction on a public ledger, tokenized carbon pools would let any participant verify credit quality before trading. The first and largest such experiment was the Base Carbon Tonne (BCT) pool, a public-ledger carbon pool (operated by Toucan Protocol) that accepted verified credits from the Verra registry (the world's largest carbon credit standard) and issued fungible units (interchangeable, each representing one tonne) at a uniform price regardless of underlying credit quality. The pool's value collapsed from approximately $7 per tonne in late 2021 to below $0.50 by mid-2023.

Industry commentary and academic analyses attributed this collapse to low-quality forest conservation credits, specifically REDD+ (a program that pays landowners to avoid deforestation)[2,6,7,8,9]. This narrative drew support from the broader literature documenting REDD+ crediting problems[4,10,11]. But it was never tested against the pool's actual composition data, which are publicly recorded on the blockchain and straightforward to query. More broadly, no study has empirically tested the dominant policy assumption that transparency prevents quality collapse.

Here, we analyze every deposit (1,187), every withdrawal (35,432), every credit token (161 individually scored, of 345 total), and every market participant (509 depositors, 28,897 redeemers) across BCT's entire operating history. The data contradict the prevailing narrative. BCT contained 69.1% renewable energy credits — predominantly grid-connected hydropower, with wind and solar, registered across China, India, Turkey, and Brazil — with near-zero additionality (meaning the projects would have been built with or without carbon revenue), and just 4.2% REDD+. A small number of accounts extracted most of the high-demand credits (1.55 million tonnes), while 9.6 million tonnes of low-quality renewables remain permanently stranded.

A comparison of identical credits across two pools with different quality screens sharpens the finding. The same credits were 100% redeemed from BCT but only 28.5% from a quality-gated pool — a contrast strongly consistent with pool design, rather than credit type, shaping asset fate. Unlike prior work that quantifies overcrediting at the project level[4,5], we trace how overcredited assets are selectively extracted once they enter a pooled market — even when quality differences are publicly observable.

We term this outcome *design-enabled adverse selection*. It is distinct from Akerlof's classical lemons problem, which requires private information about quality. Here, quality information is available to all participants but is not reflected in prices. The pool treats every deposited credit as identical: a participant who wants to buy a specific high-quality credit cannot do so. They can only buy a generic pool unit whose price reflects the pool average, not the quality of any individual credit. Uniform pricing of heterogeneous assets under voluntary participation is thus sufficient to produce market failure.

Because every transaction in this market is publicly recorded, its complete ledger supports the first credit-level forensic account of this dynamic — the kind of reconstruction that opaque off-chain carbon markets cannot supply. We treat the market as a case study of a general prediction rather than as a system of intrinsic interest; the implication for pooled markets broadly, including the $10 billion tokenized real-world asset sector and the Article 6.4 crediting mechanism (a new international carbon trading system being negotiated under the Paris Agreement), is that transparency without price differentiation appears insufficient for market function.

Results

BCT's real composition contradicts the REDD+ narrative

We extracted all 1,187 deposit records from BCT's public transaction ledger, covering 168 unique Verra VCS (Verified Carbon Standard) projects and approximately 22 million tonnes of tokenized carbon credits (see Methods).

Renewable energy credits dominated the pool (Fig. 1). Grid-connected wind, hydro, and solar projects from China and India, registered between 2008 and 2013, constituted 69.1% of total deposited tonnage. These are precisely the categories from the CDM era (the Clean Development Mechanism, a now-defunct UN crediting program, 2006–2020) that the ICVCM (Integrity Council for the Voluntary Carbon Market, the sector's main governance body) has declined to approve for its Core Carbon Principles label. Multiple independent studies document their near-zero additionality: Cames et al.[12] and Schneider et al.[13] in program-level assessments, Probst et al.[4] in a meta-analysis finding no statistically significant emission reductions from wind power CDM projects, and Calel et al.[5] estimating 52% non-additionality among Indian wind offsets. The projects would have been built regardless of carbon revenue.

As an illustrative, resolution-limited cross-check, a Sentinel-5P TROPOMI (a satellite instrument measuring atmospheric methane from space) methane-column analysis of 9 BCT waste/methane projects found 5 of 9 with site-level methane concentrations not significantly below background; its 5.5 km resolution limits project-level attribution, so we treat this only as suggestive and do not rely on it for the composition findings.

REDD+ credits, widely blamed for BCT's collapse, constituted just 4.2% of the pool — more than sixteen times smaller than the renewable share. Even combining all nature-based categories (REDD+, ARR (afforestation, reforestation, and revegetation), and IFM (improved forest management)), the total reaches only 10%, versus 69.1% for renewables alone.

Base-rate selection: BCT over-selected renewables at 1.87× the registry rate

BCT's renewable dominance could reflect the broader carbon credit supply rather than a selection effect. We obtained Verra VCS issuance data from MSCI (2024) and

Ecosystem Marketplace (2023) indicating that renewables constitute approximately 37% of total VCS issuance by volume (central estimate; robustness tested across the full sensitivity range of 26–48%).

BCT's renewable share (69.1% by tonnage; 78.5% by deposit count) yields a selection coefficient (the ratio of BCT's renewable share to the market base rate; 1 = no over-selection) of 1.87× the VCS base rate. Because deposits cluster by account (509 accounts, Gini coefficient = 0.94, where 0–1 and higher = more concentrated, effective $N = 83.5$ by HHI, the Herfindahl–Hirschman Index of market concentration), we use account-clustered inference rather than a naive binomial test. An account-level permutation test (10,000 iterations) produces $p < 0.0001$; 89.0% of accounts have majority-renewable portfolios. The selection coefficient with account-level bootstrap (resampling with replacement) 95% CI is 0.522 [0.496, 0.547] (excess renewable share above the 37% null). Under conservative assumptions (post-2008 VCS at 55%), the coefficient remains 1.43× ($p < 10^{-64}$). Over-selection becomes non-significant only at base rates exceeding 78.5%, an implausible assumption.

Conversely, REDD+ is under-represented at 0.2× the VCS base rate (BCT 4.2% vs. VCS ∼17%), directly contradicting the narrative that REDD+ credits overwhelmed the pool.

Bridge-level decomposition. Enumerating all TCO2 tokens (tokenized credits, one ERC-20 contract per Verra project) created by Toucan's factory contract reveals 369 unique tokens bridged before 2023, of which 345 (93.5%) entered BCT. By tonnage, BCT absorbed 21.98 million of 22.07 million bridged tonnes (99.6%). The 24 non-BCT tokens held a combined 86,683 tonnes (0.4%). This near-complete pass-through shows that the 1.87× over-selection was a bridge-level phenomenon: Toucan's permissionless bridge disproportionately attracted renewable energy credits from Verra, rather than depositors selectively choosing renewables from a diverse bridged universe. Pool-level selection space was essentially zero.

Price-quality feedback loop: composition predicts price collapse

The preceding sections establish what BCT contained and that its composition reflects selection. The critical question is whether quality degradation drove the price collapse.

We obtained daily BCT prices in US-dollar terms (828 observations, October 2021 – July 2024; see Methods) and merged them with daily cumulative quality metrics computed from the deposit stream ($n = 331$ overlapping days).

Correlation. BCT's price and cumulative Pool Quality Deficit (PQD, a tonnage-weighted 0–1 measure of how far a pool's credits fall below maximum quality; Methods) are strongly correlated (Pearson $r = 0.774$, $p < 10^{-66}$). In a first-differenced OLS regression (analyzing day-to-day changes rather than levels, the credible specification for non-stationary series), changes in renewable share predict price changes ($\beta = -1.8$, $p < 0.001$). Each percentage-point increase in the renewable share of deposits is associated with a $1.80 decrease in BCT's price.

Granger causality (a test of whether past values of one variable help predict another). At weekly frequency ($n = 55$), we find asymmetric bidirectional Granger causality (Fig. 2). The dominant channel runs from price to quality: price changes precede changes in pool quality (PQD) ($F = 8.40$, $p < 10^{-4}$ at lag 4), consistent with lower prices attracting lower-quality deposits. The reverse channel (quality preceding price) is significant but 2.5× weaker ($F = 3.40$, $p = 0.04$ at lag 2). This asymmetry fits the design-enabled framing: the architecture established the initial quality composition, which set a price level that then became the dominant driver of further quality deterioration. The architecture initiated the collapse; price discovery amplified it. The small sample ($n = 55$ weeks) limits statistical power for Granger tests. The first-differenced daily regression ($n = 330$, $\beta = -1.8$, $p < 0.001$) provides more powerful confirmation that quality and price co-move at the expected sign. Rolling 30-day quality metrics show no Granger causality in either direction, confirming that the feedback operates through accumulated pool composition rather than marginal deposit quality. Given small-sample limitations, we present the Granger analysis as exploratory; our primary descriptive evidence is the within-token cross-pool comparison and the first-differenced daily regression.

Redemption-side evidence. Analysis of 35,432 Transfer events from the BCT pool contract reveals that selection operated on both sides. Non-renewable credits were preferentially redeemed (Table 1): industrial gas 100%, REDD+ 99.8%, IFM 93.0%, ARR 91.3%, compared with just 3.7% of renewable credits. The tonnage-weighted mean quality of redeemed credits (38.7) exceeded that of deposited credits (31.7) by 7 points — a 22% premium. These type-level differentials (3.7% vs. 91–100%) are self-evidently large. As a result, BCT's net renewable share increased from 71% to 76% over the pool's lifetime.

The exit followed a type-level Gresham sequence (Fig. 2b). The most valuable credit types were extracted first. ARR credits exited earliest (median March 2022), followed by industrial gas (July), IFM (October), REDD+ (November), and renewable energy last (December) — perfectly ordered by off-chain market demand ($\rho = -0.74$, $n = 7$ types). Before the May 2022 market crash, redemptions targeted high-value credits (tonnage-weighted quality 42.0). Afterwards, quality dropped to 32.4, a 10-point gap reflecting selective exit while BCT's price still exceeded the off-chain value of premium credits.

Depositors and redeemers are almost entirely distinct populations: only 1.4% of 28,897 redeemer accounts also appear among the 509 depositor accounts. The top 10 redeemers account for 85.0% of extracted tonnage. Their transactions occurred in rapid automated bursts (median gap 4.6 seconds between events), consistent with programmatic execution.

This dual-margin structure — passive, architecture-driven entry on one side; active, strategic extraction on the other — is not predicted by standard adverse selection models, which assume a single population. The 99.6% tonnage pass-through shows that entry was indiscriminate. The type-specialist extraction shows that exit was deliberate. Two independent populations exploited different margins of the same mechanism.

Account forensics: who extracted and what they took

The 28,897 unique redeemer accounts are not a uniform population. Account-level analysis of the 161 scored tokens' complete transfer histories reveals a forensic pattern: the largest redeemers are type-specialist extractors who never participated in the deposit side.

Table 2. Top 5 redeemers by tonnage.

Account	Tonnes redeemed	Dominant type	Mean quality	Also depositor?
0x65a5...	651,334	Industrial gas	30.9	No
0x1b8e...	335,556	IFM	40.0	Yes
0xee4b...	195,051	REDD+	31.4	No
0x4b3e...	188,205	ARR	50.4	No
0x8556...	183,629	ARR	46.0	No

These five accounts removed 1.55 million tonnes: the vast majority of redeemed industrial gas, 37% of REDD+, and 44% of ARR deposited into BCT. Their behaviour is not uniform. Four are externally-owned wallets that redeemed nature-based credits and then re-deposited or held them, consistent with strategic arbitrage of the differential between BCT's uniform price and higher off-chain values. The largest by tonnage, however — 0x65a5. . . , 651,334 tonnes of industrial gas, 42% of the 1.55 million — is a smart contract (verified by on-chain code inspection; see Data availability) that retired every credit in the same transaction as redemption, the signature of a retirement aggregator routing credits to final use rather than an arbitrageur capturing a spread. We therefore read exit-side activity as a mix of strategic arbitrage (the four wallets) and retirement throughput (the contract). Valuing the arbitrage-consistent volume at contemporaneous off-chain prices against the BCT redemption cost yields an estimated \$3–\$9 million in captured differential (central estimate \$4–\$8 million; see Methods); this is an upper bound, as it includes volume routed through the retirement contract. Fifteen of the twenty largest redeemers never deposited anything, and each targeted a single credit type — a type-specialization inconsistent with general-purpose helpers.

Where did the extracted credits go? Tracing the immediate destination of each redeemed credit across the 20 largest extractors (2.57 million tonnes) reveals three pathways:

- **Cross-pool transfer (39.2%)**: re-deposited into a quality-screened pool (NCT, the Nature Carbon Tonne pool), a counterpart that shares identical protocol infrastructure (making it the closest available control), capturing the price differential between the unscreened pool's uniform price and the screened pool's nature-based premium.
- **Immediate retirement (34.6%)**: permanently retired (formally cancelled, never reusable as emission offsets) within seconds of extraction, indicating pre-arranged retirement pipelines. The largest extractor (0x65a5..., 651,334 tonnes of industrial gas) retired 100% of its credits in the same transaction as the redemption; an on-chain code check confirms this address is a smart contract, consistent with a retirement aggregator rather than an individual extractor.
- **Secondary market (17.0%)**: transferred to other addresses, with 6.9% re-deposited into BCT and 2.3% still held.

This cross-pool movement was not opportunistic but systematic. Of 31 credits deposited into both pools, 14 were redeemed from the unscreened pool, and all 14 follow the same temporal sequence: unscreened-pool deposit → unscreened-pool redemption → screened-pool deposit (median: 103 days, then 14 days). Not a single credit violated this ordering.

Among the 399 accounts that both deposited and redeemed (the 1.4% overlap population), quality swap analysis reveals no systematic quality arbitrage. The mean quality differential is −0.08 (effectively zero; tonnage-weighted: +3.4, driven by a few large accounts). The dual-margin mechanism operates through two distinct account populations connected only by the pool's uniform pricing: depositors who feed the pool with what the bridge produces, and extractors who take out what the off-chain market demands. We note that account-level separation does not establish entity-level independence: a single entity could operate separate deposit and redemption accounts.

Quality atlas and quality gating

BCT's Pool Quality Deficit (PQD, a 0-to-1 metric where higher values indicate worse quality; see Methods) of 0.679 places it at the bottom of the VCM quality spectrum. That spectrum spans a 10-fold range from 0.076 (DACCS, direct air carbon capture and storage) to 0.759 (grid-connected renewables) (Fig. 3). A within-pool permutation test ($z = -0.64$, $p = 0.27$) finds no evidence of within-pool selection bias; the eligible universe was uniformly low-quality. A BBB quality gate (the threshold separating credits with verified additionality from those without) would have reduced BCT's PQD to 0.405 (Fig. 6). The Toucan CHAR pool (biochar allowlist, PQD 0.221) demonstrates that category restriction prevents quality collapse (Supplementary Fig. 5).

Temporal dynamics and robustness

Temporal quality decline is a vintage-composition effect. Deposit quality declines over BCT's operating life (Spearman ρ = -0.24, n = 1,187, permutation $p < 0.0001$). This decline is entirely driven by the vintage-year component of the composite score. When the vintage dimension is excluded, the temporal trend reverses sign ($\rho = +0.24$ for renewables; $\rho = -0.01$ for all types; Supplementary Table 2). Late deposits drew from progressively older vintages, not intrinsically worse credits within vintage. This fits the design-enabled framing: the architecture attracted credit categories (CDM-era renewables) that inherently carried older vintages, but within those categories, later deposits were not of lower quality than earlier ones.

Exogenous market shock. The May 2022 cryptocurrency market crash (see Methods) provides deposit-side evidence consistent with a causal channel. Pre-shock deposits (n = 491, mean quality 32.9) versus post-shock deposits (n = 87, mean quality 29.5) show a 3.4-point quality drop (Cohen's $d = 0.49$, a standardized effect size). Volume simultaneously collapsed by 98% (14.9M to 281k tonnes). The temporal quality degradation tripled in slope ($\rho = -0.12$ pre-shock versus $\rho = -0.36$ post-shock). The exogenous price shock accelerated quality deterioration and drove volume toward zero, consistent with the price-to-quality feedback channel identified above.

Compositional shift. Renewable share rose from 67% (Q1) to 99.5% (Q4). Token diversity collapsed from 24 distinct scores to 2. Higher-quality types (IFM, Waste/Methane, ARR) ceased depositing earlier (median blocks 20.4M–21.7M) than renewables (32.6M).

Depositor concentration. Deposits were highly concentrated: 509 accounts, with the top 10 accounting for 50% of tonnage (Gini = 0.94). Large and small depositors deposited similar-quality credits (rank-biserial $r = -0.17$; the volume-weighted gap of 5 quality points does not survive permutation, $p = 0.082$).

Pool design and asset fate: a within-token cross-pool contrast

A sharper contrast on the same data is available because 14 credit tokens (1.50 million tonnes) were deposited into *both* the unscreened pool (BCT) and a quality-screened pool (NCT), so the same physical credit appears under both designs, holding quality, vintage, registry, and project type fixed. We stress what this is and is not. It is a two-pool contrast, not an experiment with 14 independent treatment assignments: BCT redeemed ∼100% of all 14 credits, so the across-token variation lies entirely on the NCT side, and the comparison therefore carries the same two-pool dependence — and the same unmodelled confounds (the pools differ in exposure timing, liquidity, and eligibility) — that led us to demote the pool-level difference-in-differences below. The result is strongly suggestive of a pool-design effect; it does not identify one, and the credits at issue are exclusively nature-based (renewables are ineligible for the screened pool), so this contrast does not bear on the renewables that dominate the pool. Reconstructing per-token deposit and redemption tonnage from raw on-chain transfer logs reproduces the aggregate redemption rates exactly (100.0% unscreened, 28.5% screened).

The unit of inference is the token ($n = 14$), not the tonne. The mean within-token redemption gap is +73.9 percentage points (unscreened minus screened; token-level bootstrap 95% CI [51, 93]) (Fig. 4). The direction is unanimous: the unscreened pool redeems more for *every* token, with the gap ranging from a few tenths of a point to a full 100 points and a median of 100. Under exchangeability of per-token signs, a paired sign test, an exact permutation test (full enumeration), and a Wilcoxon signed-rank test all return $p = 2.4 \times 10^{-4}$; this p reflects that all pairs point the same way, and is unchanged whether the one near-tied credit (BCT redemption differing from 100% by ≈1 tonne of dust in 1.5 million) is counted or dropped. The effect is not an artefact of credit-type composition — it is positive within *every* type (improved forest management +89.0, afforestation/reforestation +100.0, REDD+ +38.0 percentage points; Fig. 4). A tonne-level Beta–Binomial model yields a far tighter interval, but it treats deposited tonnes as independent Bernoulli trials when redemptions are in fact lumpy batch transactions; we therefore report it only as a descriptive summary and base inference on the token-level bootstrap and exact tests.

The estimand is the effect of pool design on the credits *admitted by both pools* — necessarily a higher-quality, nature-based subset, since the screened pool rejects renewables — not the full unscreened pool; the within-token contrast is internally clean but its external scope is this dual-eligible set. Within that set, quality, vintage, registry, and project type are held fixed by construction, so the residual threat is not credit quality but selection into the shared set: which credits came to be deposited in both pools, and in what order (for example, a credit rejected by the screened pool and then placed in the unscreened pool as a fallback). A Rosenbaum sensitivity bound (how strong a hidden confounder must be to overturn a matched-pair result) indicates the result survives an unobserved per-pair confounder that shifts the redemption odds by up to $\Gamma = 3.05$ (that odds-ratio threshold); we note that a deposit-ordering or admission-route selection rule could plausibly approach this magnitude, so we treat $\Gamma = 3.05$ as a moderate rather than a conclusive guarantee, and make the assumed causal structure explicit in the directed acyclic graph of Supplementary Fig. 3.

Generalization across pools. The mechanism is not specific to a single pool. Within the same operator, chain, time window, and depositor population, the *unscreened* pool's deposit quality degrades strongly over its operating life (Spearman $\rho = -0.439$, $p < 10^{-40}$) while the *screened* pool's does not ($\rho = -0.105$, $p = 0.088$, n.s.) — a permissionless-versus-filtered contrast that isolates screening from operator, technology, and macro conditions. Crucially, the same degradation appears in a *second, independent operator*. C3 (a different bridge protocol) encodes each tokenized credit's Verra project and vintage in its on-chain token symbol, which we recovered directly via `eth_call` (22 of 26 tokens are genuine C3 credits, four being unrelated spam tokens; 17 of 18 projects typed, after classifying five via the public registry; 75 of 92 deposits scored). On this sample C3's deposit quality also degrades over its operating life (Spearman $\rho = -0.329$, $p = 0.004$) and sits in the same low band as BCT (mean composite 31.1 versus 31.1). Because C3 credits are here scored by type alone, this slope reflects a drift toward lower-quality credit *types* over time and is independent of vintage — a cleaner signal than BCT's temporal decline, which is largely a vintage-composition effect. The mechanism is therefore not specific to Toucan: a permissionless pool from a separate operator shows the same negative temporal-quality slope. Moss MCO2, by contrast, is a single fungible basket token with no per-token credit identity, so per-token quality is genuinely undefined there and adverse selection is structurally inapplicable. We document the remaining coverage gaps explicitly (Supplementary Methods) rather than impute unobserved quality.

Pool-level difference-in-differences (descriptive). A deposit-level DiD pooling all 1,895 events (1,187 BCT + 708 NCT), with PostShock_i indicating deposits after the May 2022 market crash and standard errors clustered at the TCO2 token level, corroborates the divergence descriptively: the quality-score specification finds BCT quality falling 7.3 points more than NCT after the shock ($\hat{\beta}_3 = -7.33$, naive permutation $p < 0.0001$). We report this as descriptive only, not as a causal estimate, because BCT and NCT share 88.6% of their tokens, violating the unit-independence its inference assumes. Under a token-clustered bootstrap the standard error inflates 4.02-fold and the between-pool rank difference is indistinguishable from zero (cluster-robust $p = 0.24$). Causal identification therefore rests on the within-token design above.

Market-revealed quality hierarchy

A framework-free test asks whether credit type alone predicts extraction. A type-only prediction rule (classify high-demand categories as redeemed, all others as stranded) achieves 96.9% accuracy, 9.9 percentage points above a naive null model. The quality-grade rule (BBB+ as redeemed) achieves 91.9%. Credit type captures the dominant axis of selective redemption; the quality framework adds no predictive power beyond type classification (Supplementary Table 3). The monotonic grade-redemption pattern (B 2.4% < BB 31.0% < BBB 78.0%) reflects this: all BBB tokens are nature-based credits with strong off-chain demand, while B-grade tokens are CDM-era renewables with none. Within the 116 renewable tokens, neither vintage ($\rho = 0.112$, $p = 0.23$) nor quality grade ($p = 0.20$) significantly predicts redemption.

The paper's core findings — composition, base-rate over-selection, redemption-side extraction, and the within-token cross-pool comparison — all rest on transaction data and Verra credit-type classifications. None depend on the quality scoring framework.

In sum, 9.6 million tonnes of B-grade credits (63% of the 15.2 million tonnes covered by the 161-token scored analysis) remain unredeemed — one of the largest carbon credit graveyards in existence.

Discussion

Transparency without price differentiation does not prevent quality collapse. Every transaction was public. Every credit's quality metadata was freely available on the Verra registry and the blockchain. Yet the market collapsed exactly as economic theory predicts for markets with *hidden* quality differences. This is the central implication of the collapse we document.

The underlying principle is Gresham's Law ("bad money drives out good"), generalized by Akerlof (1970)[1] as the "lemons problem" for markets where quality is unobservable. BCT shows that the same dynamic operates even when quality *is* observable. The mechanism we term design-enabled adverse selection operates through market architecture rather than information asymmetry. Wherever heterogeneous assets are pooled at a single price, low-quality units accumulate and high-quality units are selectively extracted, regardless of whether participants can observe the difference. This complements Akerlof's framework by identifying a distinct pathway to lemons-like outcomes. Where Akerlof's mechanism requires information asymmetry, BCT shows that architectural pricing constraints alone suffice. Our within-token contrast provides credit-level descriptive evidence consistent with the lemons dynamics that Manshadi et al.[2] model theoretically, though, resting on a two-pool comparison, it does not by itself isolate the pricing-and-pooling architecture from pool-level confounds such as exposure timing, liquidity, and eligibility.

The principle can be stated compactly as a proposition: whenever heterogeneous assets are pooled at a single clearing price, with voluntary entry and exit, and the highest-quality units retain a higher value outside the pool, those units are selectively withdrawn while low-quality units accumulate — and this holds *independently of whether quality is observable*. Observability changes who profits from the spread, not whether the spread exists. The within-token result is the empirical realization of this proposition: holding the asset fixed and varying only the pooling architecture, the high-value units are extracted from the unscreened pool and retained in the screened one.

The misattribution of BCT's collapse to REDD+ credits[7,8,9] illustrates how structural quality failures can hide in plain sight. CDM-era renewable energy credits constituted 69.1% of BCT's tonnage. They suffered from near-zero additionality because the underlying projects — predominantly grid-connected hydropower, with wind and solar, across China, India, Turkey, and Brazil — were already economically competitive. Unlike REDD+ failures, which generate visible controversy over baselines (the hypothetical emission levels that would have occurred without the carbon credit project) and deforestation, the additionality failure of renewable energy credits is structurally invisible: the dams and wind farms exist, the electricity is real. Quality assurance frameworks that focus on high-profile failure modes may miss the categories that dominate pool composition.

The welfare consequences are substantial. Nearly 10 million tonnes of nominally valid offsets remain permanently stranded at a 97.6% non-redemption rate (within the 161-token scored subset). This represents an illustrative welfare gap of $100–$250 million (Monte Carlo median $146M, 90% CI $68M–$252M). Meanwhile, a small number of type-specialist accounts captured an estimated $4–$8 million (an upper bound; see Results) by selectively redeeming the credits that off-chain markets still valued — though the single largest such address is a retirement contract routing credits to final use rather than a profit-taking arbitrageur. Where Probst et al.[4] and Calel et al.[5] quantify overcrediting at the project level, we show how that overcrediting translates into participant-level extraction once credits enter a pooled market. Paradoxically, BCT's failure as a market functioned as an inadvertent quality filter. It removed 9.6 million tonnes of low-quality credits from active circulation — a permanent illiquidity event that no registry-level intervention has achieved.

The mechanism plausibly extends beyond this single market. The underlying logic — Gresham's Law under uniform pricing — applies wherever heterogeneous assets are pooled[3,14]. The same dual-margin pattern is visible, observationally, in other uniform-priced pooled markets we examined — a liquidity pool of staking derivatives and a set of non-fungible-token vaults (Supplementary Methods) — and has historical analogues in undifferentiated structured-credit and emissions-permit markets. We present these parallels as suggestive, not confirmatory: they are observational, and the on-chain cases share the same infrastructure, so they motivate rather than substitute for replication in an independent, off-chain market.

These findings carry implications for emerging carbon market architecture. They apply most directly to voluntary pooling and registry-based bundling of credits, where quality-heterogeneous units clear at a common price; the analysed pool provides a fully observable test case of what happens when quality differentiation is absent[14,15]. The Article 6.4 mechanism under the Paris Agreement differs in important respects — it is a sovereign, quantity-fixed crediting system rather than a supply-driven pool — so the lesson it inherits is the design principle (price quality, or expect its collapse) rather than a direct prediction. Recovery from a low-type equilibrium requires discontinuous intervention: even a minimal quality floor (composite score $\geq$29) would have excluded the worst half of BCT's tonnage. Gating must operate on both margins. The same 14 nature-based tokens were 100% redeemed from BCT but only 28.5% from quality-gated NCT — a within-token gap of +73.9 percentage points (token-level bootstrap 95% CI [51, 93]) — strongly consistent with pool design, not credit quality alone, shaping asset fate. These results point to three design principles for future pooled crediting mechanisms: (i) quality-differentiated pricing tiers rather than single-price pools, so that credits of different integrity levels trade at prices reflecting their climate value (for example, a three-tier structure separating high-integrity removal credits, verified avoidance credits, and legacy credits, analogous to the rating-stratified bond markets that replaced undifferentiated mortgage pools after 2008); (ii) dual-margin gating that screens both deposits and withdrawals, since exit-side extraction proved at least as damaging as entry-side quality loading (concretely, redemption fees proportional to the quality differential between the redeemed credit and the pool average would make selective extraction unprofitable); and (iii) dynamic quality floors that adjust as pool composition shifts, preventing the irreversible convergence to a low-type equilibrium (a practical implementation would trigger deposit restrictions when the pool's rolling quality deficit exceeds a threshold). Our findings provide direct empirical support for the ICVCM's Core Carbon Principles framework[16]. The emerging ecosystem of quality-differentiation mechanisms (Singapore's carbon credit ratings mandate[17], blockchain-based credibility indices[18], and quality-gating protocols[19]) must incorporate dual-margin design to be effective.

Beyond diagnosis, the same public data support a deployable early-warning. A cumulative Lemons Index computed in real time from the deposit stream (each day using only deposits already observed) crossed a 0.50 danger threshold at the pool's launch — already at 0.71 while the price was still near its \$5.91 peak — roughly nine months before the price fell below half-peak. Because this is a standing *level* signal rather than a dynamics-based predictor, it is consistent with our Granger result that price leads quality-composition changes: the diagnostic information was in the composition from inception and required no price history. A one-time composition audit on day one would have identified the risk. Paired with the quality gate (a BBB floor would have cut the pool's quality deficit from 0.679 to 0.405; Fig. 6), this converts the post-mortem into an actionable protocol: audit composition at inception, gate deposits and redemptions on quality, and re-audit as the pool evolves.

Several limitations warrant acknowledgment. Our quality scores are author-derived, but the core findings — composition, base-rate selection, type-level redemption differentials, and the within-token cross-pool comparison — rely on transaction data and Verra credit-type classifications rather than on our scoring framework. The framework is validated against CCP labels (d = 1.8; Fig. 5), commercial ratings (ρ = +0.901, n = 27; Supplementary Fig. 1), and a panel of large language models (AI systems used as independent quality raters; Fleiss' κ = 0.6; Table 4), with removal-type sensitivity checks (see Supplementary Methods). The within-token cross-pool comparison is descriptive, not a causal identification. Although it holds credit quality fixed, treatment is realized in only two pools — the same shared-unit dependence that demotes the pool-level difference-in-differences below — so the +73.9 pp gap (n = 14; token-level bootstrap 95% CI [51, 93]; sign, permutation, and Wilcoxon p = 2.4×10^{-4}) and the $\Gamma = 3.05$ bound quantify per-pair variability, not the pool-level confounds (exposure timing, liquidity, eligibility) that remain collinear with the screening label. We read it as strongly suggestive of a pool-design effect, noting that the renewables which dominate the pool are structurally excluded from this nature-based contrast. The pool-level difference-in-differences is likewise descriptive only, because shared tokens (88.6% overlap) violate the independence its inference would require (cluster-robust p = 0.24). The bridge decomposition (93.5% pass-through; 99.6% by tonnage) establishes that entry-side selection was primarily architectural rather than strategic, consistent with the design-enabled framing.

This result carries a falsifiable policy implication: any pooled crediting system that provides quality transparency without quality-differentiated pricing will experience the same collapse. The transparency reforms being implemented by ICVCM and Article 6.4 negotiators are necessary but not sufficient. If Article 6.4 pools heterogeneous credits at a single price despite its transparency requirements, the same Gresham dynamic should emerge. Mechanisms with quality-differentiated pricing tiers should not. This prediction is testable as Article 6.4 becomes operational.

The tokenized real-world asset sector now exceeds $10 billion and is replicating pool-based architecture across commodities, bonds, and receivables. BCT is the first pool to complete a full quality-collapse cycle with auditable transaction data. Its lesson is straightforward: quality-differentiated mechanism design is not an optional refinement for pooled carbon markets. It is a structural prerequisite for market integrity.

Methods

We collected all deposit transactions to the Toucan Protocol BCT pool contract on the Polygon blockchain using a Polygon RPC endpoint. We identified 1,187 deposit transactions spanning the period from BCT's launch in October 2021 to the effective cessation of new deposits in late 2022, mapping to 168 unique VCS project identifiers (345 unique TCO2 token addresses) and approximately 22 million tonnes of tokenized carbon credits. For each deposit, we recorded: transaction hash, block number and timestamp, depositor address, TCO2 token contract address, Verra VCS project identifier, vintage year, and tonnage. Project identifiers were cross-referenced against the Verra registry to obtain methodology category, country of origin, and CCP eligibility status.

Project classification. Each project was classified into one of eleven methodology categories based on the Verra registry entry: renewable energy (grid-connected wind, hydro, solar; CDM methodologies AMS-I.D, ACM0002), fossil fuel switching, waste management and methane capture, REDD+ (VM0007, VM0009, VM0015), afforestation/reforestation (ARR), industrial gas destruction, improved forest management (IFM), energy efficiency, industrial processes, agriculture, and cookstoves. Pool-level composition statistics were computed with tonnage weighting.

NCT data. We collected all 708 deposit events from the Toucan NCT pool on the same Polygon blockchain over the same block range. NCT restricts deposits to AFOLU (agriculture, forestry, and other land use) credits with vintage $\geq$2012, functioning as a minimal quality gate on project type. NCT and BCT share identical protocol infrastructure and blockchain substrate.

Price data. Daily BCT prices denominated in USDC (a stablecoin pegged to the US dollar) were obtained from the DeFi Llama API for the Polygon SushiSwap BCT-USDC pair (828 observations, October 2021 – July 2024).

Quality scoring framework

We developed a composite quality scoring framework evaluating six active dimensions — chosen to cover the primary determinants of credit integrity formalized by the Oxford Offsetting Principles and the ICVCM CCP Assessment Framework, the two most widely cited normative hierarchies for credit quality — each scored on a 0–100 scale: removal type hierarchy (weight 0.250), additionality (0.200), MRV (monitoring, reporting, and verification) grade (0.200), permanence (0.175), vintage year (0.100), and registry and methodology quality (0.075). Initial weights were derived from the Oxford Offsetting Principles[20] and the ICVCM CCP Assessment Framework[16]. A pilot Best-Worst Scaling expert panel ($n = 5$) confirmed consistency with our weights (Spearman $\rho = +0.814$). Monte Carlo perturbation (10,000 Dirichlet-sampled weight vectors) confirms 93.7% grade stability. The composite was computed as $C = \sum_i w_i \times s_i$ and mapped to letter grades: AAA ($\geq$90), AA ($\geq$75), A ($\geq$60), BBB ($\geq$45), BB ($\geq$30), B (¡30). Seven disqualifier conditions cap maximum grades regardless of composite score. The co-benefits dimension was assigned zero weight to avoid amplifying adverse selection through co-benefit narratives[6]. The Pool Quality Deficit (PQD $= 1 - \bar{C}/100$) quantifies tonnage-weighted quality degradation per pool. Full scoring details, distributional scoring model, extended scoring procedures, and sensitivity analyses are provided in Supplementary Methods.

Base-rate comparison

BCT's renewable energy share was compared against the Verra VCS base rate (central estimate 37%, sensitivity range 26–48%, from MSCI[21] and Ecosystem Marketplace reports). We define the selection coefficient as the ratio $S = f_{\text{BCT,renewable}}/f_{\text{VCS,renewable}}$ (a quantity computed for this purpose, not a borrowed named statistic), tested via exact binomial test, with account-clustered robustness (permutation, HHI-adjusted, and bootstrap tests; see Supplementary Methods).

Within-token matched-pair design (descriptive cross-pool contrast)

The within-token pool-design contrast is computed from the 14 TCO2 tokens deposited into *both* BCT (unscreened) and NCT (quality-screened). Because the same physical credit appears in both pools, quality, vintage, registry, and project type are held fixed by construction and the only difference within a pair is pool architecture; each credit is its own control. Per-token deposit and redemption tonnage for both pools were reconstructed directly from raw ERC-20 Transfer logs (deposit = wallet$\rightarrow$pool; redemption = pool$\rightarrow$wallet) cached per token, reproducing the published aggregate rates (100.0% / 28.5%) to within 0.01 percentage points and confirming the NCT per-token rates are measured, not imputed. Treating the token as the unit of analysis ($n = 14$), we tested the per-token redemption-rate difference (BCT $-$ NCT) with three exact procedures — a paired sign test, a full-enumeration permutation test over per-token sign flips, and a Wilcoxon signed-rank test — and quantified the cross-token mean gap with a percentile bootstrap over the 14 per-token differences (20,000 resamples). The exact-test p-value reflects the unanimous direction of the gap and is unchanged whether the one near-tied token (BCT redemption within ≈1 tonne of 100%) is dropped or retained. A tonne-level Beta–Binomial model (deposited tonnes as Binomial trials, uniform Beta(1,1) priors) was also fit, but it assumes independence across tonnes whereas redemptions occur in lumpy batches; it understates uncertainty and is reported only descriptively, with inference based on the token-level bootstrap and exact tests. The estimand is the pool-design effect on dual-eligible credits (those admitted by both pools), a higher-quality nature-based subset rather than the full unscreened pool. Robustness to unobserved confounding was quantified with a Rosenbaum-style sensitivity bound (the odds-ratio Γ at which the sign-test result loses significance). A mixed-effects logit with a token random intercept was also fit but exhibits quasi-complete separation (BCT ≈ 100% redemption) and is reported only as directional corroboration. The selection channel — which credits reached both pools — is bracketed by the sensitivity bound and a directed acyclic graph (Supplementary Fig. 3).

Pool-level difference-in-differences (descriptive only)

A deposit-level DiD pooling all 1,895 events (1,187 BCT + 708 NCT) was also estimated:

$$\text{quality}_i = \beta_0 + \beta_1 \cdot \text{BCT}_i + \beta_2 \cdot \text{PostShock}_i + \beta_3 \cdot (\text{BCT} \times \text{PostShock})_i + \varepsilon_i$$

where β_3 captures whether BCT's quality changed *differently* from NCT's after the May 2022 crash, with standard errors clustered at the TCO2 token level. We report it as descriptive corroboration only, not as a causal estimate: BCT and NCT share 88.6% of their tokens, violating the unit-independence its inference assumes. Under a token-clustered bootstrap the standard error inflates 4.02-fold and the between-pool rank difference is indistinguishable from zero (cluster-robust $p = 0.24$, versus naive permutation $p < 0.0001$). Causal identification therefore rests on the within-token design above.

Multi-pool generalization

To assess whether the mechanism generalises beyond a single pool, deposits to two further bridge instruments (C3 and Moss MCO2) were scored by the same methodology-archetype procedure used for BCT. For BCT and NCT we report the temporal-quality slope (Spearman ρ of composite vs deposit order) from the within-operator permissionless-versus-filtered contrast ($\rho = -0.439$ vs $\rho = -0.105$). For C3, per-token Verra project and vintage are encoded in the token symbol and were recovered on-chain via `eth_call` (22/26 tokens genuine; 17/18 projects typed, five via public-registry lookup; 75/92 deposits scored); C3's temporal-quality slope is $\rho = -0.329$ ($p = 0.004$), matching BCT's direction. Moss MCO2 is a single fungible basket token with no per-token credit identity, so a per-token quality signal is genuinely undefined and adverse selection is structurally inapplicable. Remaining coverage gaps are documented explicitly rather than imputed.

Redemption analysis

We identified redemptions from ERC-20 Transfer events where the BCT pool contract appears as the sender, yielding 35,432 events across the 161 originally scored tokens (15.2M of 22M total tonnes). Redemption rates were computed per credit type as redeemed/deposited tonnage. The within-token cross-pool comparison exploited 14 tokens deposited into both BCT and NCT to test whether the same credits experienced different fates under different pool designs. Account forensics, destination tracing, and profit quantification methods are detailed in Supplementary Methods.

Price-quality dynamics

Cumulative PQD and renewable share were merged with daily BCT prices ($n = 331$ overlapping days). A first-differenced OLS with Newey–West HAC standard errors (10 lags) tested contemporaneous co-movement. Bidirectional Granger causality was tested at weekly frequency ($n = 55$) using VAR(2) selected by AIC. We acknowledge that the small sample limits statistical power; the daily regression ($n = 330$) provides the more robust test.

Cross-domain comparison

To test whether the adverse selection pattern generalizes beyond carbon markets, we analyzed composition shifts in the Curve Finance stETH/ETH liquidity pool on Ethereum during the May 2022 market crash. Curve's stableswap treats stETH (a liquid staking derivative) and ETH as near-equivalent, a uniform-pricing mechanism analogous to BCT's 1:1 tokenization. We computed weekly net flows for each asset and defined the composition shift as the absolute difference in net flows (ETH outflow minus stETH outflow). The pre-crisis baseline was the mean weekly absolute composition shift over the 8 weeks preceding 7 May 2022. This analysis is presented as supporting evidence; the Curve comparison is observational and also consistent with a classical bank-run interpretation.

Welfare gap estimation

We estimated the welfare cost of BCT's quality degradation using a Monte Carlo simulation (10,000 iterations), chosen because multiple uncertain inputs (additionality rates, social cost of carbon, counterfactual composition) must be propagated simultaneously. Each iteration sampled: (i) additionality rates per credit type from literature-derived distributions[4,12] (renewable energy: 0–20%; REDD+: 25–75%; IFM: 30–70%; ARR: 40–80%); (ii) the social cost of carbon from a uniform distribution (\$50–\$200/tCO_2, spanning the range used by the US EPA and the Stern Review); and (iii) a counterfactual pool composition matching VCS registry base rates. The welfare gap was computed as the difference in expected climate value between the counterfactual pool and BCT's actual composition. This estimate is illustrative and conditional on both the additionality assumptions and the counterfactual specification.

Statistical inference

All p-values were corrected using the Benjamini–Hochberg FDR procedure at $\alpha = 0.05$ across the following 10 primary tests: (1) base-rate selection coefficient (binomial), (2) full-sample temporal correlation, (3) pre-shock temporal correlation, (4) within-token matched-pair sign test, (5) pool-level between-pool difference (descriptive, cluster-robust), (6) Granger quality→price, (7) Granger price→quality, (8) within-pool permutation, (9) depositor concentration Mann–Whitney, (10) redemption differential chi-squared. Headline claims were supplemented with 10,000-permutation p-values. Cluster-robust bootstrap (clustered by depositor account, 10,000 iterations) was used for deposit-level statistics. Additional inference details (account-clustered tests, CCP calibration, inter-rater reliability, rank correlations with commercial agencies) are provided in Supplementary Methods.

Data availability

All data, scoring rubrics, analysis scripts, and smart contract source code are available at https://github.com/Adeline117/carbon-neutrality under an MIT licence. Machine-readable rubrics are in JSON format. The 318-credit methodology batch dataset with per-dimension scores is included. All 345/345 BCT tokens are scored (161 original + 184 imputed). On-chain deposit data for both BCT (1,187 transactions) and NCT (708 transactions) are provided with transaction hashes enabling independent verification. The externally-owned-account versus smart-contract status of the top redeemers was checked with `eth_getCode` against a public Polygon node; results are recorded in `redeemer_contract_check.json`. LLM panel outputs for all models across 29 credits are included. Analysis scripts are pure Python with NumPy/SciPy as sole dependencies.

Code availability

All analysis code is available at https://github.com/Adeline117/carbon-neutrality under an MIT licence.

Author contributions

A.W. designed the study, collected and analyzed all data, developed the quality scoring framework, performed all statistical analyses, and wrote the manuscript. W.C. supervised the research and provided critical revisions.

Competing interests

The authors declare no competing interests.

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References

[1] Akerlof, G. A. The market for “lemons”: quality uncertainty and the market mechanism. *Q. J. Econ.* **84**, 488–500 (1970).

[2] Manshadi, V. H. et al. Offsetting carbon with lemons: adverse selection and certification in the voluntary carbon market. SSRN Working Paper (2025).

[3] Battocletti, V. et al. The voluntary carbon market: market failures and policy implications. *Colo. Law Rev.* **95**, 519–572 (2024).

[4] Probst, B. et al. Systematic assessment of the achieved emission reductions of carbon crediting projects. *Nat. Commun.* **15**, 9562 (2024).

[5] Calel, R. et al. Do carbon offsets offset carbon? *Am. Econ. J. Appl. Econ.* **17**, 1–40 (2025).

[6] Berg, F. et al. The market for voluntary carbon offsets. SSRN Working Paper (2025).

[7] Bosshard, C. et al. Blockchain-based voluntary carbon market: strategic insights into network structure. *Front. Blockchain* **8**, 1603695 (2025).

[8] Ballesteros-Rodriguez, D., De-Lucio, J. & Sicilia, G. Tokenized carbon credits in voluntary carbon markets: the case of KlimaDAO. *Front. Blockchain* **7**, 1474540 (2024).

[9] Jirasek, M. KlimaDAO: a crypto answer to carbon markets. *J. Organ. Des.* **12**, 271–283 (2023).

[10] Trencher, G. et al. Demand for low-quality offsets by major companies undermines climate integrity of the voluntary carbon market. *Nat. Commun.* **15**, 6863 (2024).

[11] West, T. A. P. et al. Action needed to make carbon offsets from forest conservation work for climate change mitigation. *Science* **381**, 873–877 (2023).

[12] Cames, M. et al. How additional is the Clean Development Mechanism? Oko-Institut Report (2016).

[13] Schneider, L. Assessing the additionality of CDM projects: practical experiences and lessons learned. *Clim. Policy* **9**, 242–254 (2009).

[14] World Bank. State and Trends of Carbon Pricing 2025. World Bank Group (2025).

[15] Garcia, A. & Sanford, L. On the potential for strategic behaviour in jurisdictional REDD+. *Proc. Natl Acad. Sci. USA* **123**, e2531612123 (2026).

[16] Integrity Council for the Voluntary Carbon Market. Core Carbon Principles, Assessment Framework, and Assessment Procedure. ICVCM (2023).

[17] Singapore National Environment Agency. Carbon rating panel: appointment of BeZero, Calyx Global, and Sylvera. NEA Regulatory Notice (2025).

[18] Gao, H. O. & Liu, X. A blockchain-based carbon registry platform for credible climate action in transportation. *npj Clim. Action* **5**, 23 (2026).

[19] [Companion paper] ERC-CCQR: A composable quality rating standard for tokenized carbon credits. Manuscript in preparation.

[20] Allen, M. et al. The Oxford Principles for Net Zero Aligned Carbon Offsetting. University of Oxford (2020).

[21] MSCI. State of integrity in the global carbon-credit market. MSCI ESG Research (2024).

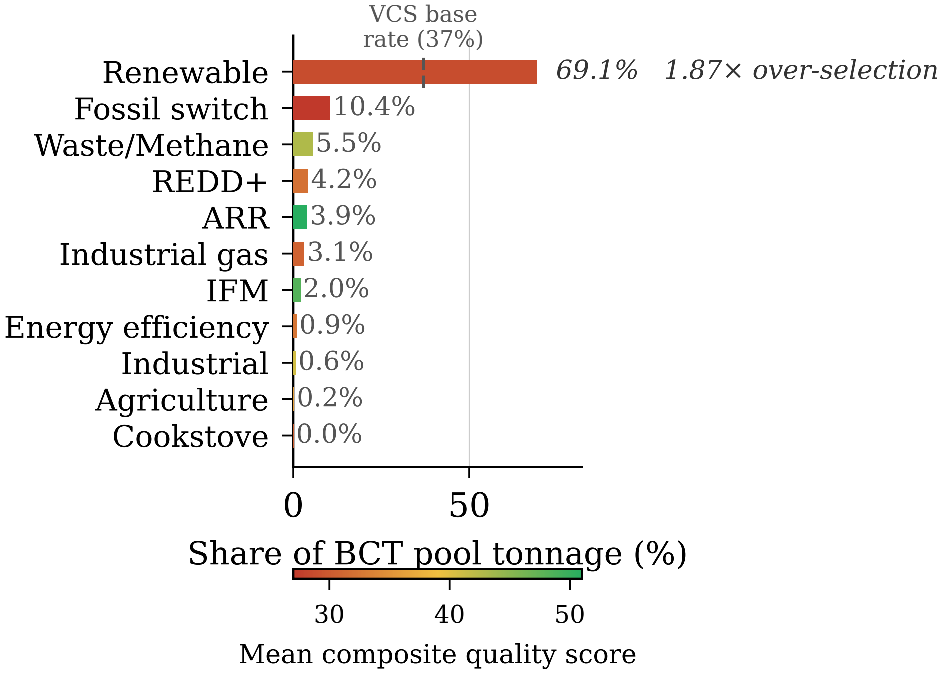

Fig. 1 BCT pool composition by methodology category, weighted by deposited tonnage. Renewable energy credits constitute 69.1% of total deposited tonnage; REDD+ constitutes just 4.2%.

BCT Price Trajectory and Pool Quality Composition

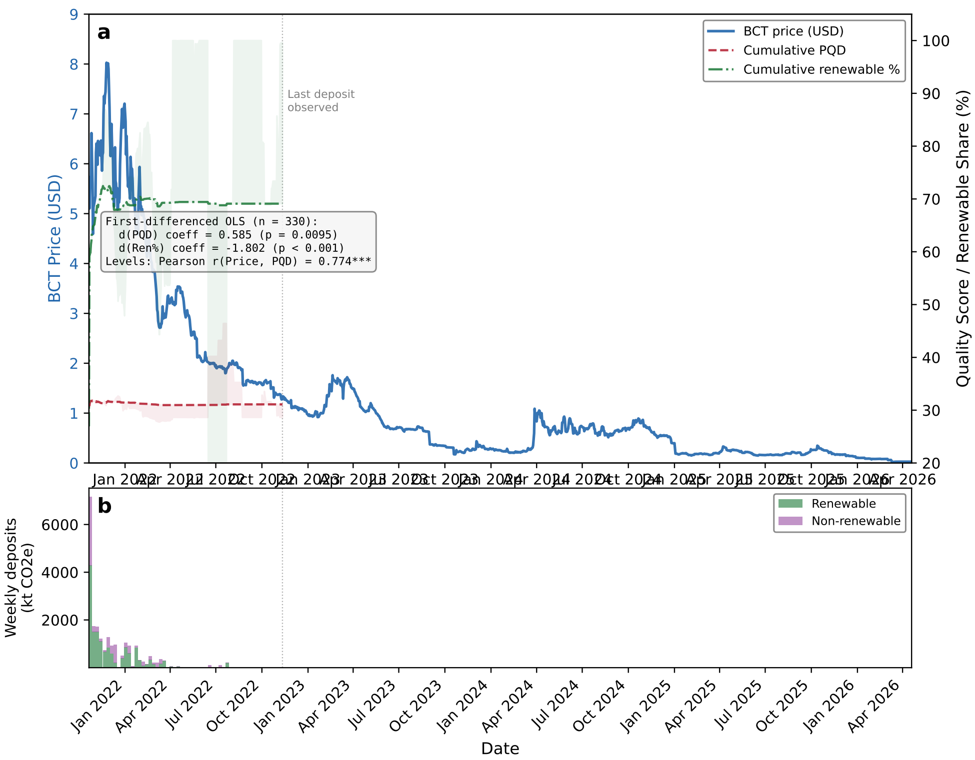

Fig. 2 Price-quality feedback loop: bidirectional Granger causality between BCT price and cumulative quality composition.

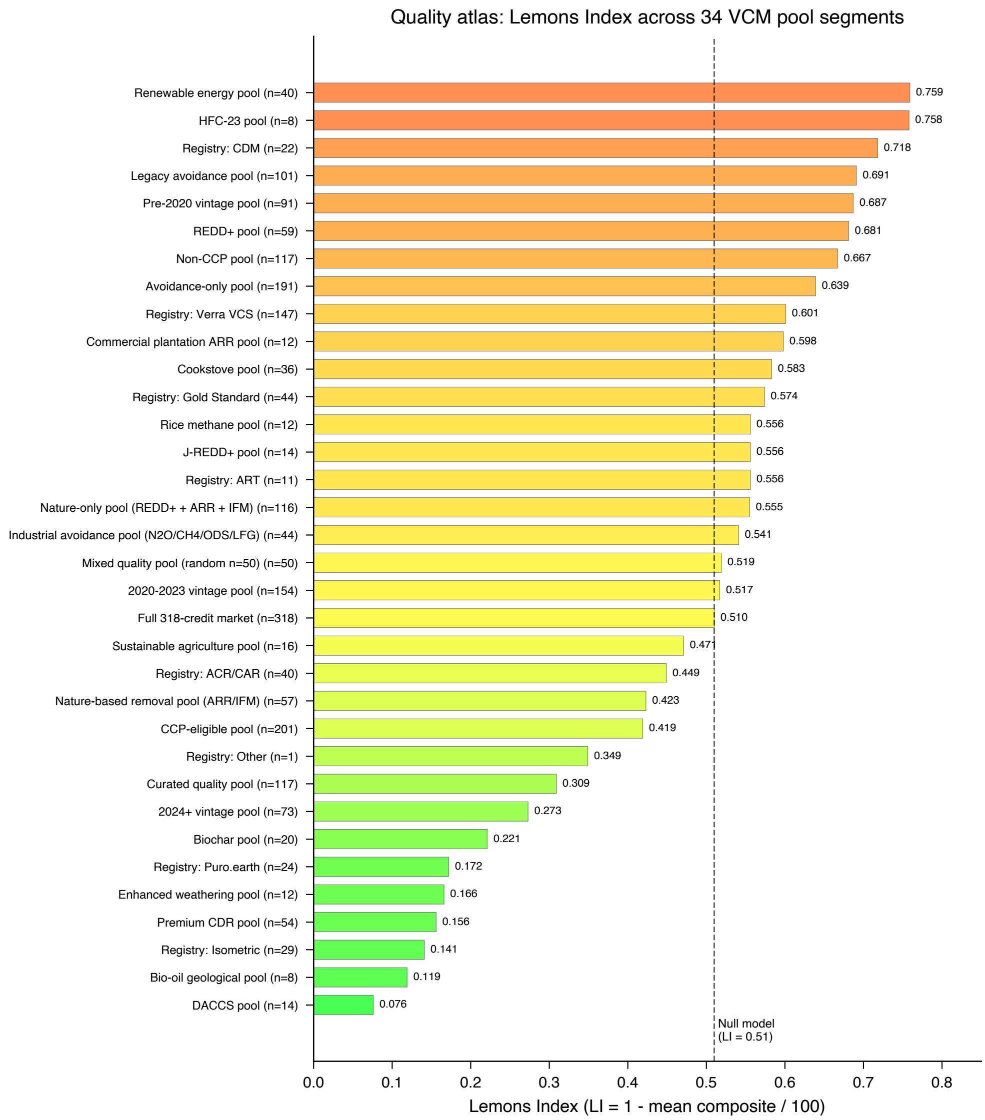

Fig. 3 Quality atlas: 34-segment PQD spectrum with null model baseline. PQD ranges from 0.076 (DACCS) to 0.759 (grid-connected renewables).

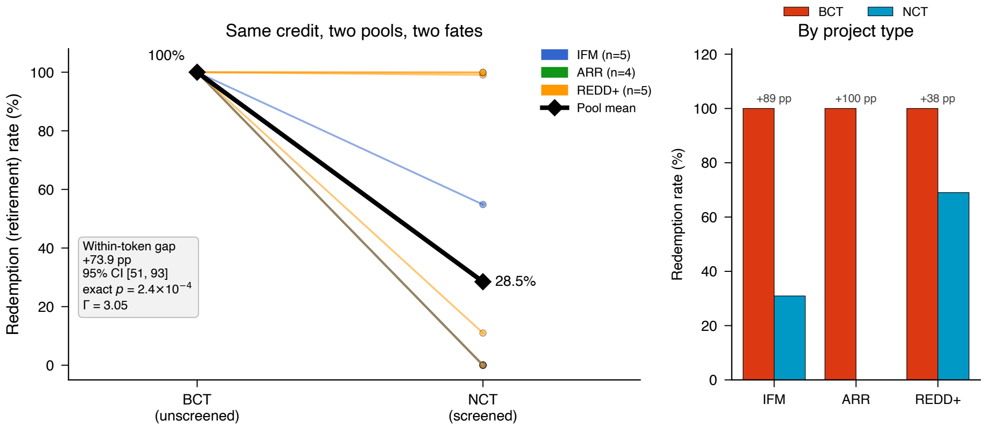

Fig. 4 Within-token cross-pool contrast: redemption of identical credits under two pool designs (descriptive; a two-pool comparison, not a causal identification). (**a**) Per-token slope plot of redemption (retirement) rate for the 14 carbon credit tokens deposited into *both* the unscreened BCT pool and the quality-screened NCT pool, coloured by project type; the bold black line traces the pool means (100.0% redeemed from the unscreened pool versus 28.5% from the screened pool). Across the 14 pairs (the unit of inference) the mean within-token redemption gap is +73.9 percentage points (token-level bootstrap 95% CI [51, 93]); the direction is unanimous, so exact sign, permutation, and Wilcoxon tests all return $p = 2.4 \times 10^{-4}$ whether or not the one near-tied credit is included. (**b**) Redemption rate by project type (tonnage-weighted): the gap is positive within every type (IFM +89, ARR +100, REDD+ +38 pp). The result is robust to unobserved confounding up to a Rosenbaum bound of $\Gamma = 3.05$. Depositor-level temporal, base-rate, and difference-in-differences robustness panels are reported in Supplementary Fig. 3.

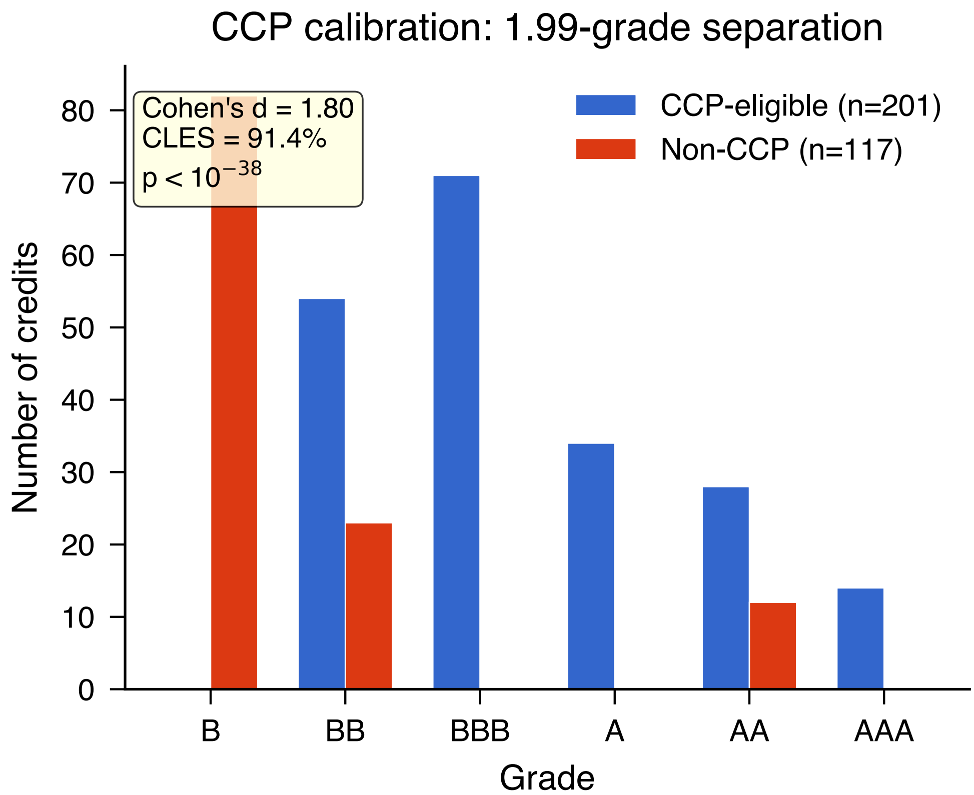

Fig. 5 CCP calibration: quality score distributions for CCP-eligible versus non-CCP credits. Cohen's $d = 1.87$.

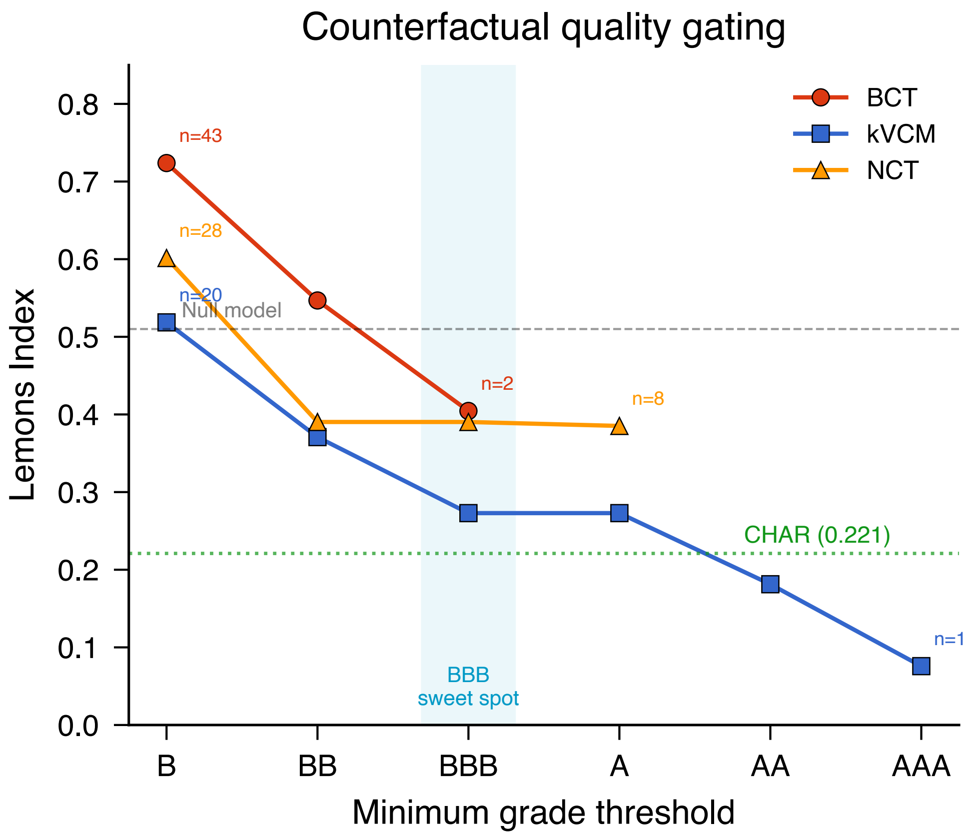

Fig. 6 Counterfactual quality gating: PQD reduction under progressive grade thresholds for four pools.